07 – SQL: Structured Query Language

A language to communicate with a database

Jérôme Landré – jerome.landre@ju.se

OUTLINE

0) Recall

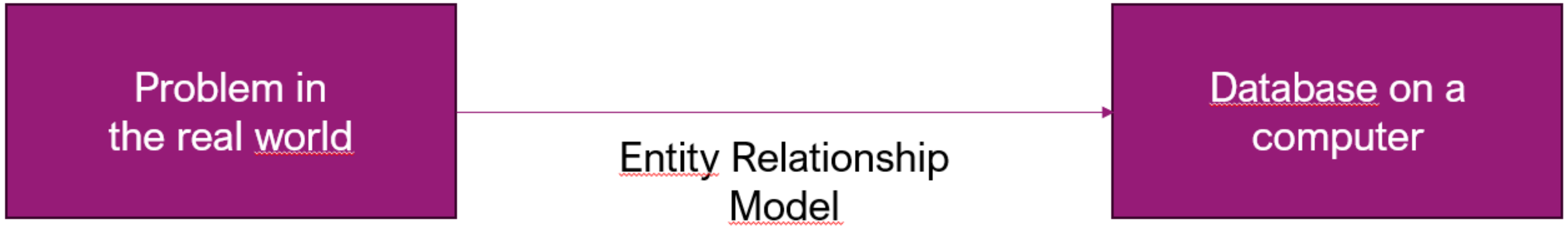
1) What is SQL?

2) Data Definition Language (DDL)

3) Data Manipulation Language (DML)

4) SQL Joins or not…

0) RECALL

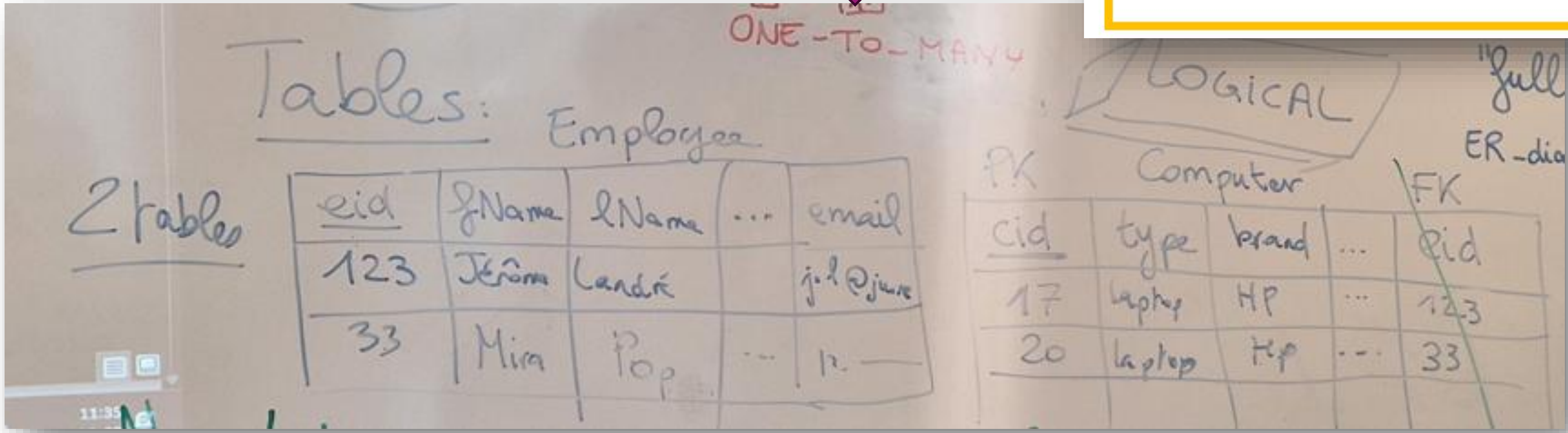

Problem in the real world
Entity Relationship Model
Database on a computer

0) RECALL

- To create a Relational Database, the traditional way is to use the **Entity-Relationship (ER) Model** which has three levels:

  - The **conceptual** model: an Entity-Relationship Diagram (ERD) representing the entities, attributes, relationships, and cardinalities
  - The **logical** model: a list of tables
  - The **physical** model: the organization of the information on the computer hard drive (files)

- Today, we enter the **LOGICAL MODEL**: **TABLES**



RULES:

1) Any **entity** becomes a **table**.
2) **Look** at the **cardinalities (max only)**:
    a. a **one-to-many** cardinality makes the **PK** on the 1 side become a **FK** into the table on the * side.
    b. a **many-to-many** relation becomes a **full table** with **PK** being the two **FK** of the tables in the relation.

1) WHAT IS SQL?

SQL (Structured Query Language) is a **standardized programming language** used for managing and manipulating relational databases [1][2][3]. It allows users to store, retrieve, update, and delete data in database management systems [5].

SQL was developed in the 1970s and has since become the **global standard** for relational database management systems (RDBMS)[4]. It is widely used by database administrators, developers, and data analysts for various tasks, including:

1. Executing queries to retrieve data
2. Inserting new records into databases
3. Updating existing records
4. Deleting records from databases
5. Creating new databases and tables
6. Setting permissions on database objects

SQL is particularly useful for handling structured data and managing relationships between different data entities [3]. It operates on a **declarative model**, meaning **users describe what they want to achieve rather than specifying how to do it** [4].

1) WHAT IS SQL?

Key features of SQL include:

- **Standardization**: SQL became an ANSI standard in 1986 and an ISO standard in 1987 [2][3].
- **Versatility**: It can be used across various database systems, including MySQL, Oracle, and Microsoft SQL Server [5].
- **Scalability**: SQL can handle growing data needs effectively [4].
- **Data manipulation**: It excels at processing and analyzing large datasets [4].

While SQL has a standardized core, different database systems may have their own dialects or extensions, such as T-SQL for MS SQL Server and PL/SQL for Oracle [4].

Citations:
[1] https://www.geeksforgeeks.org/what-is-sql/
[2] https://www.w3schools.com/sql/sql_intro.asp
[3] https://en.wikipedia.org/?title=SQL
[4] https://www.ibm.com/think/topics/structured-query-language
[5] https://www.techtarget.com/searchdatamanagement/definition/SQL
[6] https://www.atlassian.com/data/sql
[7] https://www.reddit.com/r/AskProgramming/comments/16o4ijx/this_may_sound_a_bit_stupid_but_what_exactly_is/
[8] https://www.youtube.com/watch?v=27axs9dO7AE
Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?

Structured Query Language (SQL) is the relational databases

data definition language (**DDL**) and **data manipulation language** (**DML**).

It is a **language**, you must **learn** its **syntax** to be able to use it!

1) WHAT IS SQL?

Data Definition Language (DDL) is a subset of SQL used to define and manage the structure of database objects [1][2]. It provides commands for creating, modifying, and deleting database elements such as tables, indexes, views, and sequences [2][3].

- Key Aspects of DDL:

1. **Purpose**: DDL is used to describe, comment on, and label database objects, as well as impose or drop constraints on tables [2].
2. **Common Commands**: The most frequently used DDL commands are CREATE, ALTER, and DROP [1]. For example:
   - **CREATE**: Used to create new database objects like tables or indexes [1].
   - **ALTER**: Modifies existing database structures [1].
   - **DROP**: Removes database objects [1].
3. **Scope**: DDL deals with the structure of the database, not the data itself[1]. It affects the schema of the database rather than the content [3].
4. **Immediate Effect**: When a DDL statement is executed, it takes effect immediately in the database [1].
5. **User Access**: Typically, DDL commands are not used by general application users but by database administrators or developers with appropriate permissions [1].

DDL plays a crucial role in database management by providing the tools necessary to define and maintain the structural integrity of databases, ensuring efficient data storage and retrieval [5].

Citations:
[1] https://www.techtarget.com/whatis/definition/Data-Definition-Language-DDL
[2] https://www.secoda.co/glossary/what-is-a-data-definition-language-ddl
[3] https://en.wikipedia.org/wiki/Data_definition_language
[4] https://www.getdbt.com/blog/guide-to-ddl
[5] https://www.lenovo.com/us/en/glossary/what-is-ddl/
[6] https://www.techopedia.com/definition/1175/data-definition-language-ddl
[7] https://celerdata.com/glossary/data-definition-language-ddl
Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?

Data Manipulation Language (DML) is a subset of SQL used for managing and manipulating data within a database. It focuses on interacting with the actual data stored in database tables, allowing users to perform CRUD (Create, Read, Update, Delete) operations [1][3].

- Key DML Commands:
1. **SELECT**: Retrieves data from one or more tables [5]. Data Retrieval: Fetching specific data based on certain criteria.
2. **INSERT**: Adds new records to a table [3][5]. Data Entry: Inserting new records into database tables.
3. **UPDATE**: Modifies existing data in a table [3][5]. Data Modification: Updating existing records with new information.
4. **DELETE**: Removes records from a table [3][5]. Data Removal: Deleting unnecessary or outdated records.

- Characteristics of DML:
    - DML statements are executed immediately and can be undone with a rollback statement [2].
    - They are typically used by end-users, applications, or systems interacting with the database [2].
    - DML operations affect the data within tables, not the database structure itself [4].

DML plays a crucial role in database management by providing the tools necessary for efficient data handling and retrieval, ensuring that databases remain up-to-date and relevant for various applications and business needs [3][7].

Citations:
[1] https://www.datasunrise.com/knowledge-center/dml-data-manipulation-language/
[2] https://appmaster.io/blog/difference-between-ddl-and-dml
[3] https://satoricyber.com/glossary/dml-data-manipulation-language/
[4] https://www.javatpoint.com/ddl-vs-dml
[5] https://en.wikipedia.org/wiki/Data_manipulation_language
[6] https://stackoverflow.com/questions/2578194/what-are-ddl-and-dml
[7] https://staragile.com/blog/data-manipulation-language
[8] https://www.scaler.com/topics/ddl-dml-dcl/

Answer from Perplexity: pplx.ai/share

1) WHAT IS SQL?

2) DATA DEFINITION LANGUAGE (DDL)

How to create a table?

```
-- SQL comment, create a new table
CREATE TABLE Person (
    pId INTEGER PRIMARY KEY ASC,
    pFName TEXT,
    pLName TEXT,
    pBirthDate TEXT,
    pEmail TEXT,
    pPhone TEXT
);
```

How to **delete** a table?

```
-- SQL comment, delete table (and all content!)
DROP TABLE Person;
```

Expression Affinity	Column Declared Type
TEXT	"TEXT"
NUMERIC	"NUM"
INTEGER	"INT"
REAL	"REAL"
BLOB (a.k.a "NONE")	"" (empty string)

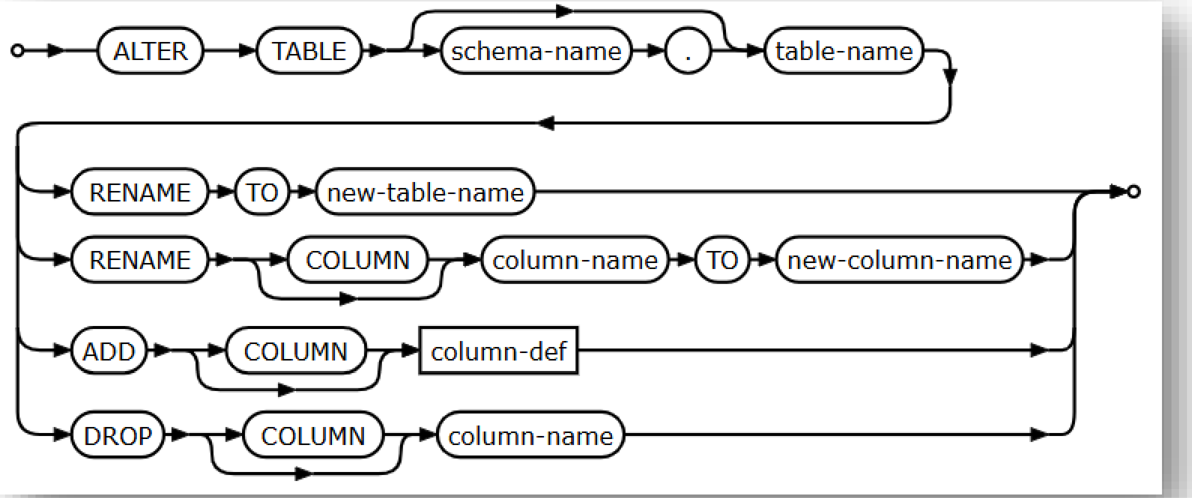
We NEED **datatypes**!!!

Source: https://sqlite.org/lang_createtable.html

2) DATA DEFINITION LANGUAGE (DDL)

How to **modify/alter** a table?

-- SQL comment, alter a column type
ALTER TABLE Person
RENAME COLUMN pPhone to phone;



Source: https://sqlite.org/lang_altertable.html

3) DATA MANIPULATION LANGUAGE (DML)

C R U D

Create	Read	Update	Delete
INSERT	SELECT	UPDATE	DELETE

3) DATA MANIPULATION LANGUAGE (DML)

How to insert a new row?

```
-- Insert a new row
INSERT INTO Person
(pFName, pLName, pEmail, pPhone)
VALUES
(”Jérôme”, ”Landré”, ”j.l@ju.se”, ”070605040302”)
```

How to delete a row?

```
-- Delete a row
DELETE FROM Person
WHERE pId=123;
```

3) DATA MANIPULATION LANGUAGE (DML)

How to update/modify a row?

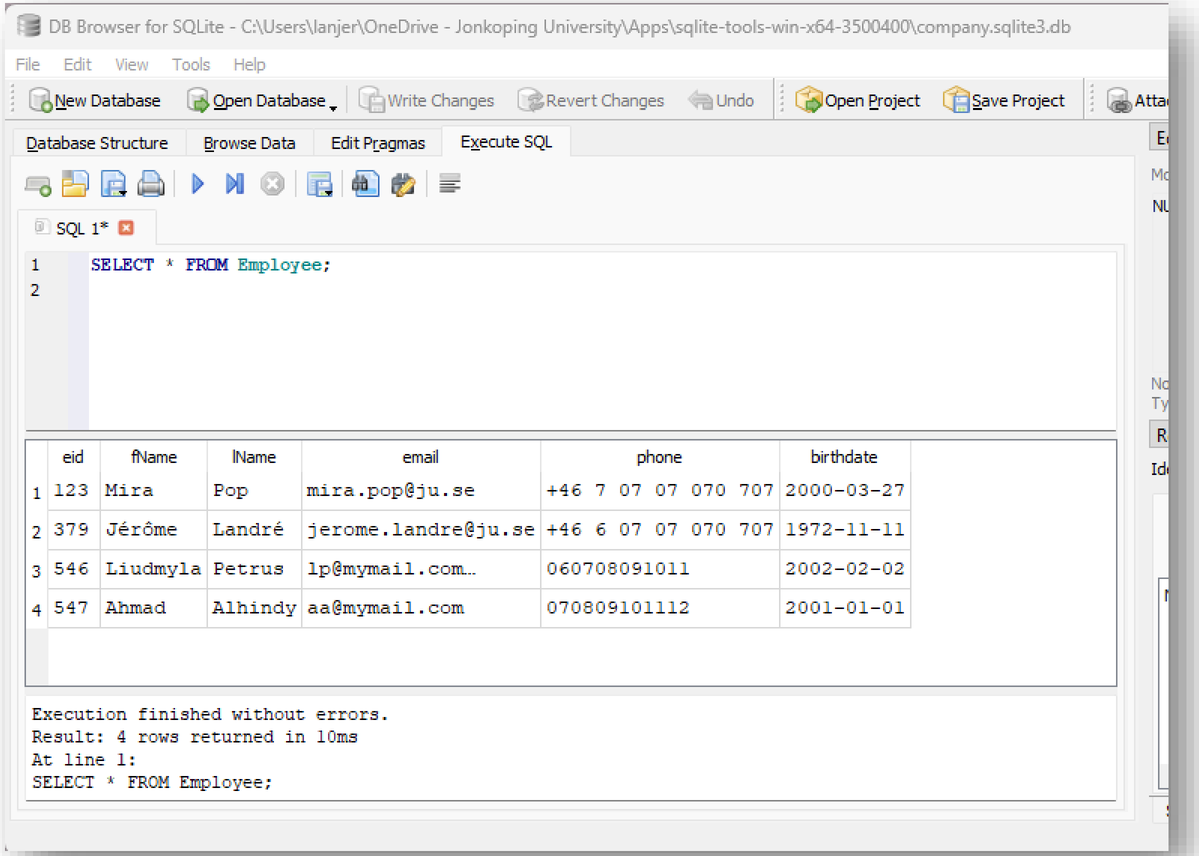
```
-- Update/Modify a row
UPDATE Person
SET pFname="Jérôme", pPhone="0607080910"
WHERE pId=123;

-- Update/Modify many rows
UPDATE Person
SET salary=salary*1.01  -- increase by 1%
WHERE pBirthDate>"2000-01-01";  -- young persons
```

How to read a row/rows?

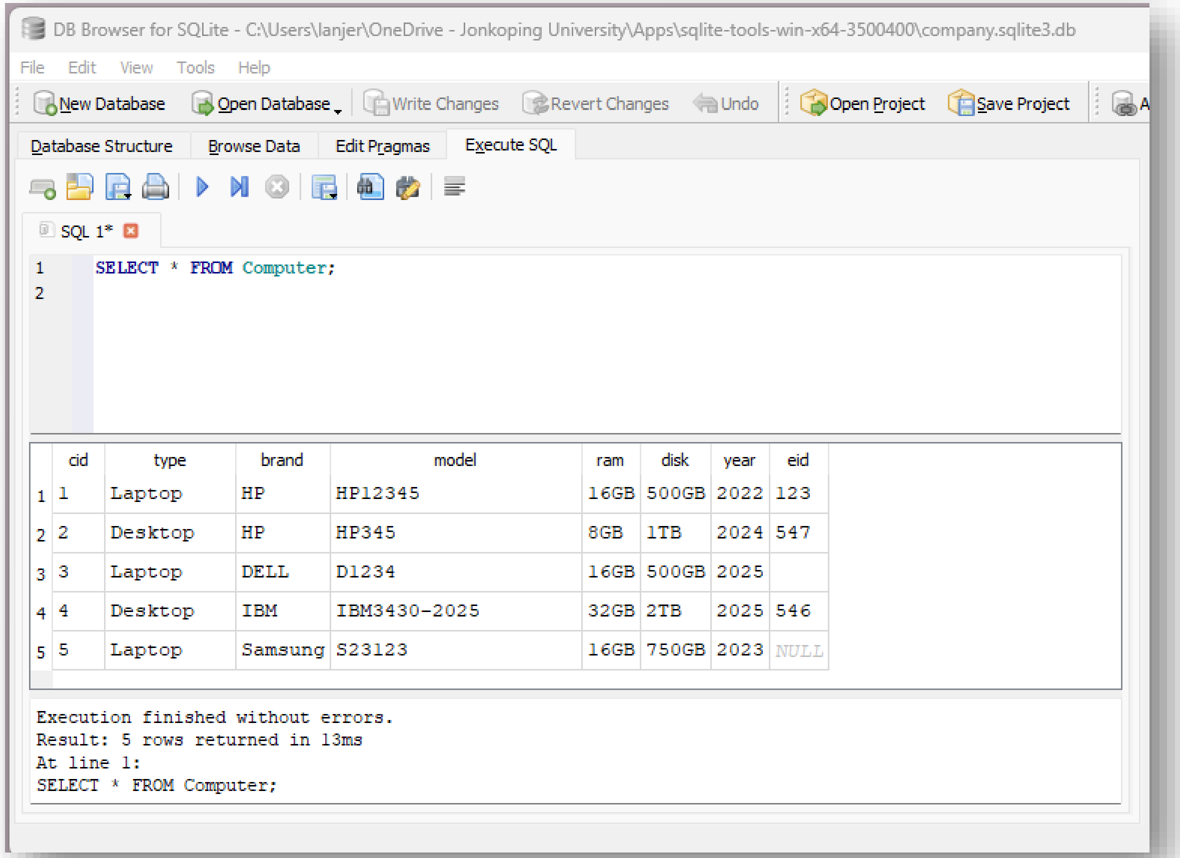
```
-- Read one row
SELECT * FROM Person
WHERE pId=123;

-- Read many rows
SELECT * FROM Person
WHERE pBirthDate>"2000-01-01";
```

5) SQL JOINS OR NOT…

- If the information you want to extract is present only in one table, no need to join tables, easy:



	eid	fName	lName	email	phone	birthdate
1	123	Mira	Pop	mira.pop@ju.se	+46 7 07 07 070 707	2000-03-27
2	379	Jérôme	Landré	jerome.landre@ju.se	+46 6 07 07 070 707	1972-11-11
3	546	Liudmyla	Petrus	lp@mymail.com...	060708091011	2002-02-02
4	547	Ahmad	Alhindy	aa@mymail.com	070809101112	2001-01-01



	cid	type	brand	model	ram	disk	year	eid
1	1	Laptop	HP	HP12345	16GB	500GB	2022	123
2	2	Desktop	HP	HP345	8GB	1TB	2024	547
3	3	Laptop	DELL	D1234	16GB	500GB	2025	
4	4	Desktop	IBM	IBM3430-2025	32GB	2TB	2025	546
5	5	Laptop	Samsung	S23123	16GB	750GB	2023	NULL

5) SQL JOINS OR NOT…

- If the information you want to extract is present **only in one table**, no need to join tables, easy:

```
SELECT * FROM Computer
WHERE type='Laptop';
```

	cid	type	brand	model	ram	disk	year	eid
1	1	Laptop	HP	HP12345	16GB	500GB	2022	123
2	3	Laptop	DELL	D1234	16GB	500GB	2025	
3	5	Laptop	Samsung	S23123	16GB	750GB	2023	NULL

```
SELECT * FROM Computer
WHERE year<=2023;
```

	cid	type	brand	model	ram	disk	year	eid
1	1	Laptop	HP	HP12345	16GB	500GB	2022	123
2	5	Laptop	Samsung	S23123	16GB	750GB	2023	NULL

```
SELECT * FROM Employee
WHERE birthdate>='2001-01-01';
```

	eid	fName	lName	email	phone	birthdate
1	546	Liudmyla	Petrus	lp@mymail.c...	060708091011	2002-02-02
2	547	Ahmad	Alhindy	aa@mymail.com	070809101112	2001-01-01

```
SELECT * FROM Employee
WHERE email LIKE '%ju.se';
```

	eid	fName	lName	email	phone	birthdate
1	123	Mira	Pop	mira.pop@ju.se	+46 7 07 07 070 707	2000-03-27
2	379	Jérôme	Landré	jerome.landre@ju.se	+46 6 07 07 070 707	1972-11-11

5) SQL JOINS OR NOT…

- When the information you want to get is **across multiple tables**, we need to use a JOIN **between these tables to retrieve the data**
- For instance, what are the computers of employee number 547?

```
1  SELECT * FROM Employee;
2
3
4
```

	eid	fName	lName	email	phone	birthdate
1	123	Mira	Pop	mira.pop@ju.se	+46 7 07 07 070 707	2000-03-27
2	379	Jérôme	Landré	jerome.landre@ju.se	+46 6 07 07 070 707	1972-11-11
3	546	Liudmyla	Petrus	lp@mymail.com...	060708091011	2002-02-02
4	547	Ahmad	Alhindy	aa@mymail.com	070809101112	2001-01-01

```
1  SELECT * FROM Computer;
2
```

	cid	type	brand	model	ram	disk	year	eid
1	1	Laptop	HP	HP12345	16GB	500GB	2022	123
2	2	Desktop	HP	HP345	8GB	1TB	2024	547
3	3	Laptop	DELL	D1234	16GB	500GB	2025	
4	4	Desktop	IBM	IBM3430-2025	32GB	2TB	2025	546
5	5	Laptop	Samsung	S23123	16GB	750GB	2023	NULL

5) SQL JOINS OR NOT…

JÖNKÖPING UNIVERSITY
School of Engineering

- When the information you want to get is **across multiple tables**, we need to use a JOIN **between these tables to retrieve the data**
- For instance, what are the computers of employee number 547? **Using JOIN**

```sql
SELECT * FROM Employee
INNER JOIN Computer
ON Computer.eid=Employee.eid
WHERE Employee.eid=547;
```

	eid	fName	lName	email	phone	birthdate	cid	type	brand	model	ram	disk	year	eid
1	547	Ahmad	Alhindy	aa@mymail.com	070809101112	2001-01-01	2	Desktop	HP	HP345	8GB	1TB	2024	547

5) SQL JOINS OR NOT…

- When the information you want to get is **across multiple tables**, we need to use a JOIN **between these tables to retrieve the data**
- For instance, what are the computers of Jérôme? **Using JOIN**

```
SELECT * FROM Employee
INNER JOIN Computer
ON Computer.eid=Employee.eid
WHERE Employee.fName='Jérôme';
```

eid	fName	lName	email	phone	birthdate	cid	type	brand	model	ram	disk	year	eid

5) SQL JOINS OR NOT…

- When the information you want to get is **across multiple tables**, we need to use a JOIN **between these tables to retrieve the data**
- For instance, what are the computers of Mira? **Using JOIN**

```
SELECT * FROM Employee
INNER JOIN Computer
ON Computer.eid=Employee.eid
WHERE Employee.fName='Mira';
```

	eid	fName	lName	email	phone	birthdate	cid	type	brand	model	ram	disk	year	eid
1	123	Mira	Pop	mira.pop@ju.se	+46 7 07 07 070 707	2000-03-27	1	Laptop	HP	HP12345	16GB	500GB	2022	123

5) SQL JOINS OR NOT…

- When the information you want to get is **across multiple tables**, we need to use a JOIN **between these tables to retrieve the data**
- For instance, List the computers assigned to a person with the firstname, lastname, type, brand and year of the computers? **Using JOIN**

```
SELECT fName, lName, type, brand, year FROM Employee
INNER JOIN Computer
ON Computer.eid=Employee.eid;
```

	fName	lName	type	brand	year
1	Mira	Pop	Laptop	HP	2022
2	Ahmad	Alhindy	Desktop	HP	2024
3	Liudmyla	Petrus	Desktop	IBM	2025

ANY QUESTIONS?